

Multiplicative Negative K-Theory and Quantum Cellular Automata

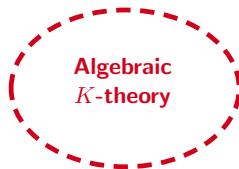
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Algebraic Structures in Topology 2026 (Graduate Workshop)

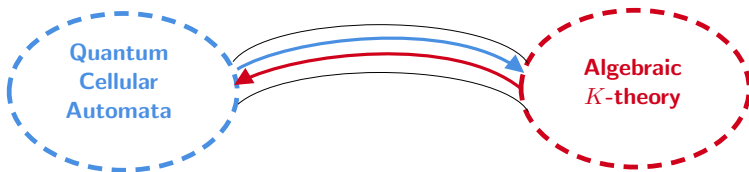
July 12th, 2026

This talk is about two seemingly disjoint subjects:

- 1 Quantum Cellular Automata (QCA) from quantum computing.
- 2 Algebraic K -theory from homotopy theory.



... and a **bridge** between the two subjects.



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2 Algebraic K -theory

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The Heisenberg Picture for Quantum Mechanics

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Let V be a:

- finite dimensional complex Hilbert space (i.e., $\mathbb{C}^n$)

The **Heisenberg picture** for quantum mechanics treats a quantum system as a non-commutative algebra of observables

$$\mathcal{A} := \text{End}(V).$$

$\mathcal{A}$ has an adjoint operation $*$ coming from the Hermitian inner product.

Symmetries of quantum systems are then modeled as $*$ -preserving automorphisms of $\mathcal{A}$, which is precisely

the projective unitary group $PU(n)$.

Many-Particle Quantum Systems

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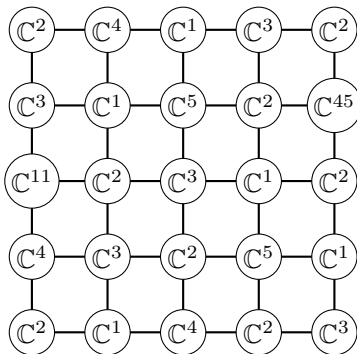
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The discussions prior can be viewed as **single-particle quantum mechanics**:

$$V = \mathbb{C}^n$$



Often times, one would like to model quantum systems on **multiple (and even infinitely many) particles**.



What is Quantum Cellular Automata

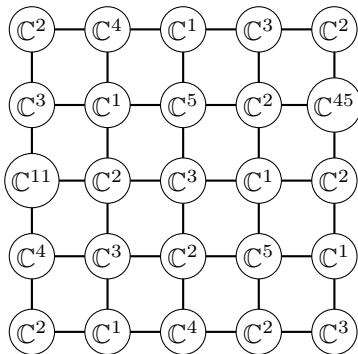
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In the many-body case over a lattice X ,

- ① The algebra of observables $\mathcal{A}(X, q)$ becomes a colimit of matrix algebras.
- ② Quantum cellular automata (QCA) is the physically appropriate automorphism group of $\mathcal{A}(X, q)$.

What is a QCA, More Formally.

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Throughout this talk, we will let:

- ① R be a commutative ring with unity.
- ② X be a nice metric space¹, such as $\mathbb{Z}^n$ with the L^∞ -metric or lattices in Lie groups.

The placement of vector spaces (or free R -modules) on X is given by the following.

Definition:

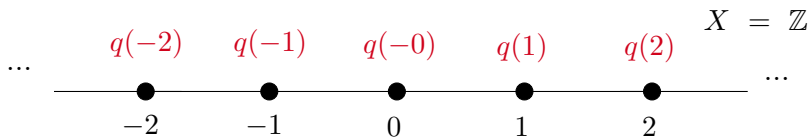
A quantum spin system is a set function

$$q : X \rightarrow \mathbb{Z}_{\geq 1} = \{1, 2, 3, \dots\}$$

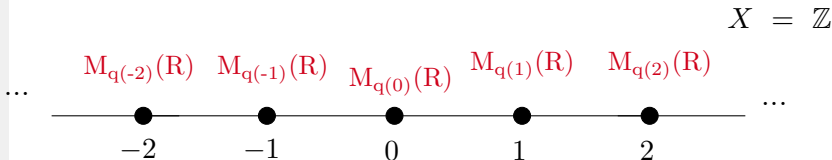
such that for each $y \in X$ and $r \geq 0$,

$$B_r(y) \cap \{x \in X : q(x) > 1\} \text{ is finite.}$$

¹We assume X contains countable, coarsely dense subspace that is uniformly discrete, locally finite metric space of bounded geometry.



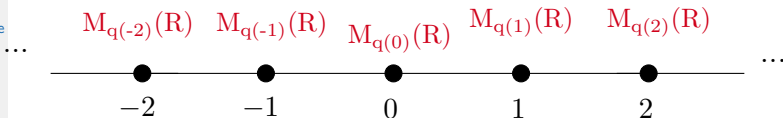
Since we only care about the endomorphisms, morally we might as well think of a quantum spin system as the picture:



Here $M_n(R)$ is the **noncommutative R -algebra of $n \times n$ matrices over R** .

Informal:

The **algebra of observables** $\mathcal{A}(X, q)$ is the “**infinite tensor product**” of $M_{q(x)}(R), x \in X$.



More precisely,

$$\mathcal{A}(X, q) := \operatorname{colim}_{\text{finite subsets } S \subseteq X} \underbrace{\bigotimes_{s \in S} M_{q(s)}(R)}_{=:\mathcal{A}(S, q)},^2$$

where for $S \subseteq T$ finite subsets,

$$\mathcal{A}(S, q) \rightarrow \mathcal{A}(T, q)$$

is given by $- \otimes \operatorname{id}_{\mathcal{A}(T-S, q)}$, tensoring with the identity operator on $T - S$.

²Algebras like these are examples of what are called [supernatural algebra](#) [Bar-On et al., 2023].

A **quantum cellular automaton** α is an R -algebra automorphism

$$\alpha : \mathcal{A}(X, q) \rightarrow \mathcal{A}(X, q)$$

of **finite spread**, meaning that there exists some $\ell > 0$ such that for all $\gamma \in M_{q(x)}(R)$, $\alpha(\gamma) \in \mathcal{A}(D_\ell(x), q)$.

In other words, a QCA is an “**automorphism that moves components away in a distance that is uniformly bounded**”.

Definition:

The group of QCAs over $\mathcal{A}(X, q)$ is denoted $\mathcal{Q}(X, q)$.

Example of QCA

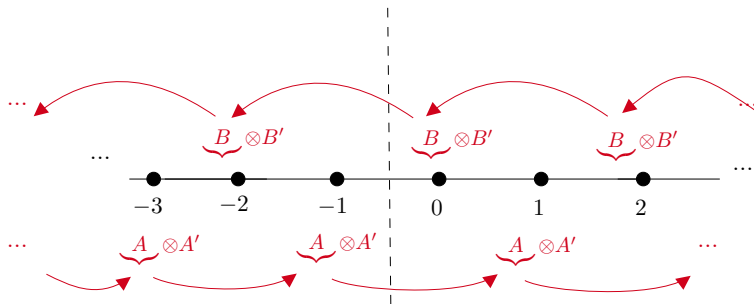
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A QCA α for a pair (A, B) of Azumaya R -algebras, B' and A' are tensor complements such that $A \otimes A' \cong M_n(R)$ and $B \otimes B' \cong M_m(R)$.

Not An Example of QCA

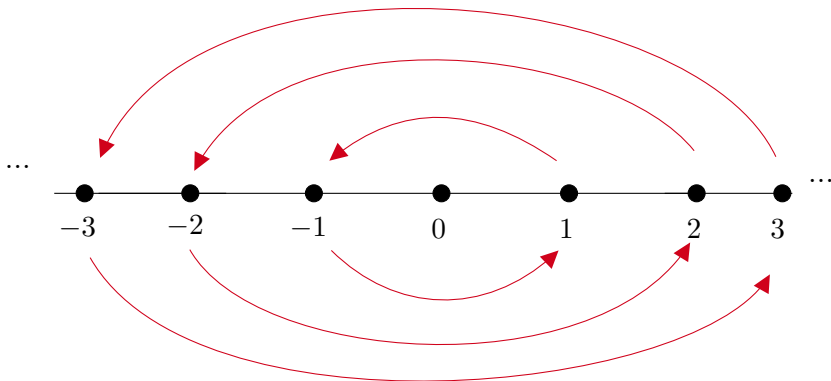
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Here $q : X \rightarrow \mathbb{Z}_{>0}$ is constant, the antipodal map swaps the tensor factors.

Example of 2D QCA

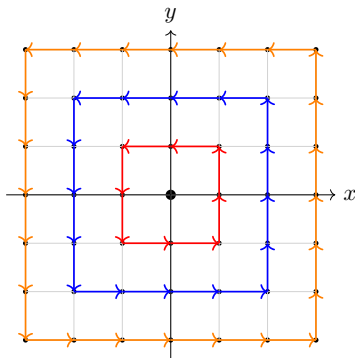
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Annular shifts on the plane. This has spread 1.

We can **stabilize** the QCA groups from earlier.

- 1 For **quantum spin systems** $q, r : X \rightarrow \mathbb{Z}_{\geq 1}$, we say $q \leq r$ if

$$q(x) \text{ divides } r(x), \quad \forall x \in X.$$

- 2 For $q \leq r$, we can define a map

$$\mathcal{Q}(X, q) \rightarrow \mathcal{Q}(X, r)$$

by pointwise tensor product with the **identity matrix** of dimension $r(x)/q(x)$.

- 3 The total QCA group is

$$\mathcal{Q}(X) := \operatorname{colim}_q \mathcal{Q}(X, q).$$

The Classification Theory of QCAs

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Definition:

The **circuit subgroup** of $\mathcal{Q}(X)$ is the commutator subgroup $\mathcal{C}(X) = [\mathcal{Q}(X), \mathcal{Q}(X)]^3$.

A central question in QCA is:

Question:

How can we classify QCAs up to circuits?

³The circuit subgroup has physical meaning and was defined differently, which is omitted for the sake of time. In [Ji and Yang, 2026b], we showed they are equivalent in most cases of interest.

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Question:

How can we classify QCAs up to circuits?

The groups $\mathcal{Q}(X)^{ab} := \mathcal{Q}(X)/\mathcal{C}(X)$ are infamously difficult to compute! In the (unitary) case over $\mathbb{C}$:

- ① $\mathcal{Q}(\mathbb{Z}^1)^{ab} = \mathbb{Q}_{>0}$ GNVW index [Gross et al., 2012].
- ② $\mathcal{Q}(\mathbb{Z}^2)^{ab} = 0$ by a theorem of Michael Freedman and Matthew Hasting [Freedman and Hastings, 2020].
- ③ Jeongwan Haah recently obtained a proof that $\mathcal{Q}(\mathbb{Z}^3)^{ab}$ contains a copy of $\mathbb{Z}/2\mathbb{Z}$.
- ④ The higher groups are difficult in general.

It has been conjectured [Haah, 2021] that $\mathcal{Q}(\mathbb{Z}^3)^{ab}$ is related/equal to the Witt group of braided fusion $\mathbb{C}$ -linear categories.

Why Homotopy Theory?

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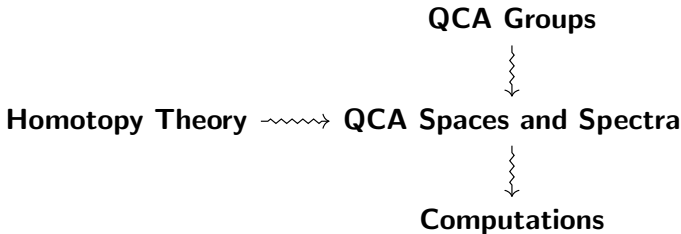
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Influenced by Kitaev's conjecture of Ω -spectrum for invertible states, there has been a **folklore conjecture** in the field that the **classification of QCA should be governed by an Ω -spectrum**.

This has also been mentioned explicitly in [Long and Else, 2025, Tu et al., 2025].



The QCA Conjecture

For each spatial dimension d , there is a **space-level construction** $\mathbf{Q}(\mathbb{Z}^d)$ of the QCA classification group with a canonical isomorphism

$$\pi_0 \mathbf{Q}(\mathbb{Z}^d) \cong \mathcal{Q}(\mathbb{Z}^d) / \mathcal{C}(\mathbb{Z}^d).$$

Moreover, these spaces assemble naturally into an **Ω -spectrum** $\mathbf{QCA} = \{\mathbf{Q}(\mathbb{Z}^d)\}_{d \geq 0}$, satisfying certain expected properties.

The QCA Conjecture

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Theorem ([Ji and Yang, 2026b])

The QCA conjecture is true over any ring R . Furthermore, for any nice metric space X , there is an analogous Ω -spectrum of the form

$$\mathbb{QCA}(X) : \mathbf{Q}(X), \mathbf{Q}(X \times \mathbb{Z}), \mathbf{Q}(X \times \mathbb{Z}^2), \dots$$

- 1 The QCA conjecture is the **special case** when $X = *$ is a single point.
- 2 When $R = \mathbb{C}$, this can be adapted to the unitary case at the beginning to give an **unitary QCA spectrum**.

Our strategy uses a subject called **algebraic K -theory**.

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What is Algebraic K -theory?

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Question:

What is Algebraic K -theory?

On a high level, K -theory is a general methodology / abstraction of the process of:

- ① Breaking an object apart.
- ② Assembling the pieces together.
- ③ Studying structural properties from the additive data.

Conceptual Illustration

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For a ...

Field k

The objects are ...

Finite Dimensional
 k -Vector-Spaces

We decompose by ...

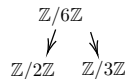
Direct Sums



Ring R

Finitely Generated
Project R -Modules

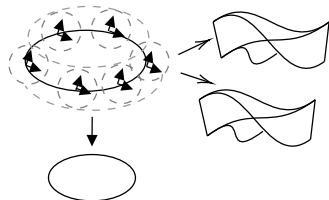
Direct Sums



*Topological
Space X*

Finite Rank Vector
Bundles on X

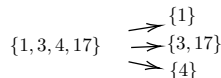
Whitney Sums



*Symmetric
Monoidal
Category $\mathcal{C}$*

Objects in $\mathcal{C}$

The tensor
product $\otimes$



What is Algebraic K -Theory

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Algebraic K -theory is a functor

$$\text{Symmetric Monoidal Categories} \xrightarrow{K(-)} \text{Spc}.$$

For a symmetric monoidal category $\mathcal{S}$, its i -th K -theory is:

$$K_i(\mathcal{S}) := \pi_i K(\mathcal{S}).$$

Example:

For a ring R , let $\text{Proj}(R)^{f.g.}$ denote the symmetric monoidal groupoid of finitely generated projective R -modules, with direct sum as the operation.

The K -theory of R is $K(R) := K(\text{Proj}(R)^{f.g.})$.

Negative K -Theory

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The definition above accounts for the **higher K -theory** of R .

Bass considered instead the lower K -groups that he called **negative K -theory**.

- ① These are groups $K_{-i}(R)$ for $i \geq 0$.
- ② They are useful invariants in detecting whether or not a ring is smooth, among other applications.

Definition:

Classically, they are defined as follows:

- ① $K_{-1}(R) := \operatorname{coker}(K_0(R[t]) \oplus K_0(R[t^{-1}]) \rightarrow K_0(R[t^{\pm 1}]))$.
 - ② Since the construction is functorial, one iteratively defines
- $$K_{-(n+1)}(R) := \operatorname{coker}(K_{-n}(R[t]) \oplus K_{-n}(R[t^{-1}]) \rightarrow K_{-n}(R[t^{\pm 1}]))$$

Question:

Can the groups $K_{-i}(R)$ be realized as the **negative homotopy groups** of a non-connective delooping of the space $K(R)$?

[Pedersen and Weibel, 1985] gave a concrete way to construct this, which will look **quite familiar**.

The Pedersen-Weibel Construction

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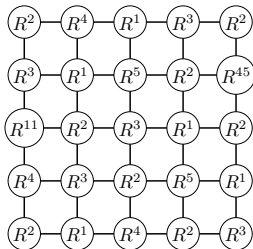
For each $i \geq 0$, the **Pedersen-Weibel bounded category** $\mathcal{C}_i(R)$ is a **symmetric monoidal category** whose:

- **Objects** are $\bigoplus_{x \in \mathbb{Z}^i} R^{q(x)}$ for $q : \mathbb{Z}^n \rightarrow \mathbb{Z}_{\geq 0}$.
- **Morphisms** are R -linear isomorphisms

$$\phi_{x,y} : R^{q(x)} \rightarrow R^{q(y)}$$

such that $\phi_{x,y} \equiv 0$ if $d(x,y) > \ell$.

- **Symmetric Monoidal** under **pointwise direct sum**.



The Pedersen-Weibel Delooping

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Theorem ([Pedersen, 1984])

For $i \geq 0$, $K_{-i}(R) = K_1(\mathcal{C}_{i+1}(R))$.

The construction of **PW bounded category** in fact works for **any additive category**.

Theorem ([Pedersen and Weibel, 1985], Stated Informally.)

After a suitable idempotent completion procedure, the spaces $K(\mathcal{C}_i(R))$ for $i \geq 0$ form a non-connective delooping of $K(R)$, denoted $\mathbb{K}(R)$.

Clearly Pedersen and Weibel's construction works for **any nice metric space X** , this was studied in a follow-up paper [Pedersen and Weibel, 1989].

This perspective has been developed into what is called **controlled K -theory** [Carlsson and Goldfarb, 2011].

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For the input (X, R) , we define a **QCA category** $\mathbf{C}(X; R)$ whose:

- Objects are $\mathcal{A}(X, q)$.
- Morphisms of R -algebra isomorphisms of **finite spread**.⁴
- Symmetric monoidal under **pointwise tensor product** $\otimes$.

Example:

- When $X = *$, $\mathbf{C}(*; R)$ is the symmetric monoidal category of matrix rings $M_n(R)$, under tensor product.

$$K_0(\mathbf{C}(*; R)) \cong \mathbb{Q}_{>0}.$$

⁴Importantly, $\mathbf{C}(X; R)$ is not usually skeletal

Structural Properties of the QCA Category

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Theorem ([Ji and Yang, 2026b])

- ① When $\text{Spec } R$ is connected,
 $K_0(\mathbf{C}(X; R)) \cong \text{CH}_0(X; \bigoplus_{\mathbb{N}} \mathbb{Z})$, the 0-th coarse homology group of X in coefficient $\bigoplus_{\mathbb{N}} \mathbb{Z}$.
- ② For $i \geq 1$, $K_0(\mathbf{C}(\mathbb{Z}^i; R)) = 0$.
- ③ $K_1(\mathbf{C}(X; R)) = \mathcal{Q}(X)/\mathcal{C}(X)$.
- ④ $K_1(\mathbf{C}(\mathbb{Z}^1; R)) = K_0(\text{Az}_R, \otimes)$, where Az_R is the category of Azumaya R -algebras.
- ⑤ $K(\mathbf{C}(X; R)) \cong K_0(\mathbf{C}(X; R)) \times B\mathcal{Q}(X)^+$.

The QCA Space

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The QCA classification group being at π_1 motivated⁵ us to take a loop space.

Definition

The **QCA space** of $(X; R)$ is

$$\mathbf{Q}(X; R) := \Omega(K(\mathbf{C}(X; R))_1).$$

From here, our main theorem was:

Theorem ([Ji and Yang, 2026b])

There is a canonical delooping

$$\mathbf{Q}(X; R) \cong \Omega \mathbf{Q}(X \times \mathbb{Z}; R).$$

⁵This was suggested to us by Tomer Schlank.

Theorem ([Ji and Yang, 2026b])

There is a canonical delooping

$$\mathbf{Q}(X; R) \cong \Omega \mathbf{Q}(X \times \mathbb{Z}; R).$$

There are certain challenges in proving this that Pedersen-Weibel delooping did not encounter:

- ① The category $\mathbf{C}(X; R)$ is **Not Additive**.
- ② **Idempotent completion** does not really make sense for $\mathbf{C}(X; R)$. For direct sums, there is an obvious projection

$$\pi_A : A \oplus B \rightarrow A.$$

The analog of such projection **does not really make sense** for

$$\pi_A : A \otimes B \rightarrow A.$$

- ③ We cannot **split an element** into its individual parts at each point $x \in X$.

Theorem ([Ji and Yang, 2026b])

There is a canonical delooping

$$\mathbf{Q}(X; R) \cong \Omega \mathbf{Q}(X \times \mathbb{Z}; R).$$

Proof: The proof required a lot of soul-searching in [algebraic topology](#) and [noncommutative algebra](#).

- 1 In the proof of the main theorem, we showed something stronger that

$$\Omega K(\mathbf{C}(X \times \mathbb{Z}; R)) = K(\text{Invertible } R\text{-Subalgebras of } X)$$

which generalized a theorem of [Haah, 2023] when we take π_0 on both sides.

- 2 Using this, we could actually identify

$$\mathbf{Q}(\mathbb{Z}^1; R) = K(Az_R, \otimes).$$

Thus, the QCA spectrum $\mathbf{QCA}(*; R)$ can also be seen as a non-connective delooping of $K(Az_R, \otimes)$.

- 3 For each X , $\mathbf{Q}(X)/\mathbf{C}(X) \cong H_2(\mathbf{C}(X \times \mathbb{Z}^1); \mathbb{Z})$.

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Algebraic K -theory	Quantum Cellular Automata
Free R -Modules R^n	Matrix R -Algebras $M_n(R)$
Projective R -Modules	Azumaya R -Algebras
Direct Sum $\oplus$	Tensor Product $\otimes$
Reduced $\tilde{K}_0(R)$	Brauer Group $\text{Br}(R)$
Negative K -theory	QCA Groups
The spectrum $\mathbb{K}(R)$	The spectrum $\text{QCA}(*; R)$
Controlled K -theory	$\text{QCA}(X; R)$

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Direct Sum $\oplus$	Tensor Product $\otimes$
Reduced $\widetilde{K}_0(R)$	Brauer Group $\text{Br}(R)$
Negative K -theory	QCA Groups
The spectrum $\mathbb{K}(R)$	The spectrum $\text{QCA}(*; R)$
Controlled K -theory	$\text{QCA}(X; R)$

Thematic Questions:

What can we say about **QCA groups** using **techniques from K -theorists?**

What can we say about **multiplicative K -theory** using **techniques from the quantum physicists?**

If you are interested to see more, check out our papers:

Initial paper [resolving the QCA Conjecture](#):

-  Ji, M. and Yang, B. (2026b). [Quantum Cellular Automata: The Group, the Space, and the Spectrum](#)
(<https://arxiv.org/abs/2602.16572>)

Follow-up paper on applications to [linearization](#) and [onsiteability](#):

-  Ji, M. and Yang, B. (2026a). [K-Theoretic Obstructions to Linearizing QCA Representations](#)
(<https://arxiv.org/abs/2606.19657>)



Bar-On, T., Gilat, S., Matzri, E., and Vishne, U. (2023).
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Multiplicative
Negative
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and
Quantum
Cellular
Automata

Mattie Ji,
joint work
with Bowen
Yang

Quantum
Cellular
Automata
(QCA)

Algebraic
 K -theory

From
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QCA



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